TOC - Retest Paper - CS354TA

PART-A

1.1 $r = 0^* + 1^*$, $s = 01^* + 10^* + 1^*0 + (0^*1)^*$, $t = 1^*(01)^*0^*$

a) str. of min. length not in language corresponding to R.E 't' (excluding ϵ).

$\Rightarrow$ 001, 011

b) corresponding to 's' not in 't'

$\Rightarrow$ 001, 011,

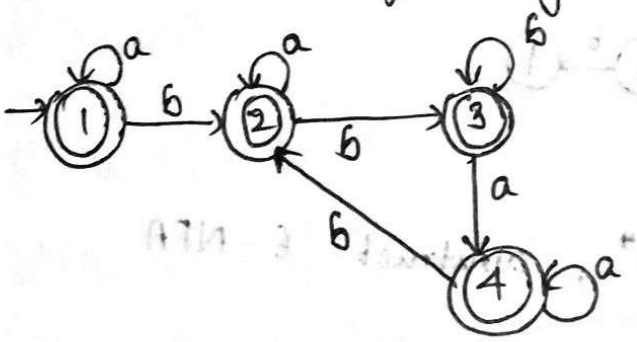
c) Both r and s

$\Rightarrow \epsilon, 0, 1,$

d) neither 'r' and 's'

$\Rightarrow$ 101, 010

1.2 Minimize the following DFA



When all states in a DFA are accepting, the minimized DFA will always have a single state with self-loops for all inputs

$\therefore$

where $q_f = \{1, 2, 3, 4\}$

$\therefore$ No of states = 1

1.3 Deterministic Finite Automata

- A DFA is defined by the 5-tuple:-

$$\{Q, \Sigma, q_0, F, \delta\}$$

where

$Q \Rightarrow$ a finite set of states

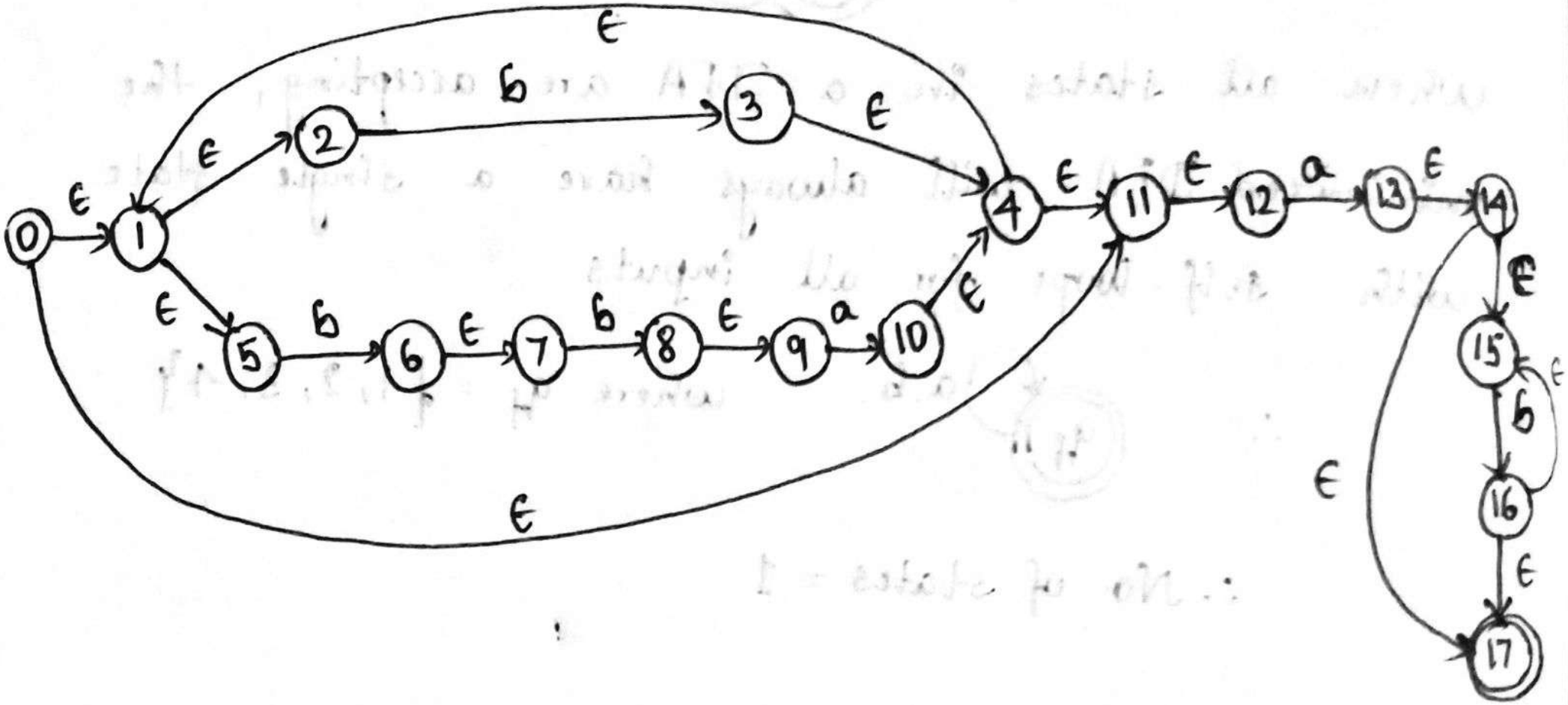
$\Sigma \Rightarrow$ a finite set of i/p symbols [alphabet]

$q_0 \Rightarrow$ start state

$F \Rightarrow$ set of accepting states

$\delta \Rightarrow$ transition function which is a mapping between $Q \times \Sigma \Rightarrow Q$

1.4 $R = (b + bba)^* ab^*$, construct ϵ-NFA

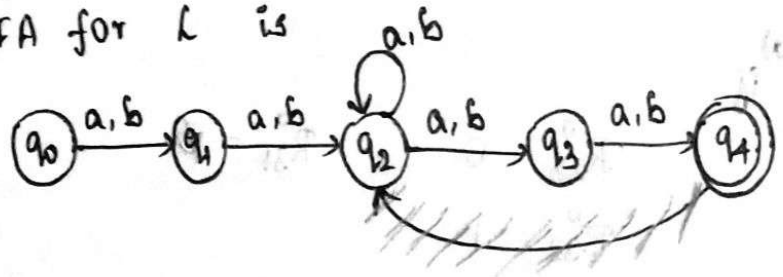


1.5 $L = \{vwv : v, w \in \{a,b\}^*, |v| = 2\}$ is regular

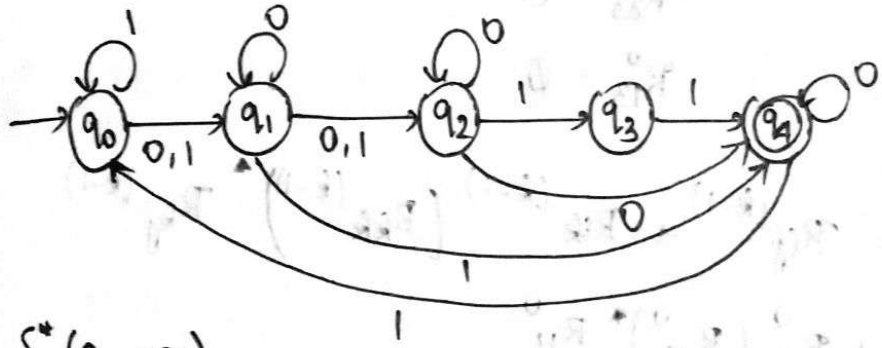
$\Rightarrow$ If we can come up with

- DFA
- NFA
- Regular Expression, for some Language L, then L is regular

$\therefore$ NFA for L is



1.6



$\delta^*(q_0, 001)$

$\delta^*(q_0, \epsilon) = q_0$

$\delta^*(q_0, 0) = \delta(\delta^*(q_0, \epsilon), 0) = \delta(q_0, 0) = q_1$

$\delta^*(q_0, 00) = \delta(\delta^*(q_0, 0), 0) = \delta(q_1, 0) = \{q_1, q_2\}$

$\delta^*(q_0, 001) = \delta(\delta^*(q_0, 00), 1) = \delta(q_1, 1) \cup \delta(q_2, 1)$

$= \{q_2, q_4\} \cup \{q_3\}$

$= \{q_2, q_4, q_3\}$

1.7 Two decision properties of Regular Languages

i) Emptiness test

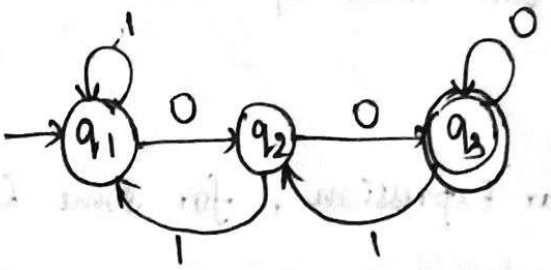
ii) Membership test

iii) Finiteness test

PART - B

2) DFA

	0	1
$\rightarrow q_1$	q_2	q_1
q_2	q_3	q_1
$* q_3$	q_3	q_2



⇒ transition diagram

a) R.E of $R^{(0)}_{ij}$

$R^0_{11} = 1$ $\quad R^0_{23} = 0$ $\quad R^0_{31} = \emptyset$

$R^0_{12} = 0$ $\quad R^0_{32} = 1$

$R^0_{21} = 1$ $\quad R^0_{33} = 0$

$R^0_{22} = \epsilon$ $\quad R^0_{13} = \emptyset$

b) $R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} \left(R_{kk}^{(k-1)}\right)^* R_{kj}^{(k-1)}$

$R^1_{11} = R^0_{11} + R^0_{11} (R^0_{11})^* R^0_{11}$

$= 1 + 1(1)^* 1 = 11^*$

$R^1_{12} = R^0_{12} + R^0_{11} (R^0_{11})^* R^0_{12}$

$= 0 + 1(1)^* 0 = 1^* 0$

$R^1_{21} = R^0_{21} + R^0_{21} (R^0_{11})^* R^0_{11}$

$= 1 + 1(1)^* 1 = 11^*$

$R^1_{22} = R^0_{22} + R^0_{21} (R^0_{11})^* R^0_{12}$

$= \epsilon + 1(1)^* 0 = \epsilon + 1(1)^* 0$

$R^1_{23} = R^0_{23} + R^0_{21} (R^0_{11})^* R^0_{13}$

$= 0 + 1(1)^* \emptyset = 0$

$R^1_{32} = R^0_{32} + R^0_{31} (R^0_{11})^* R^0_{12} = 1 + \emptyset(\cdots)$

$= 1$

$$R_{33}^{1} = R_{33}^{0} + R_{31}^{0}\left(R_{11}^{0}\right)^{*} R_{13}^{0}$$

$$= 0 + \emptyset\,(\cdots)$$

$$= 0$$

$$R_{13}^{1} = R_{13}^{0} + R_{11}^{0}\left(R_{11}^{0}\right)^{*} R_{13}^{0}$$

$$= \emptyset + 1\,(1)^{*}\,\emptyset \quad = \emptyset$$

$$R_{31}^{1} = R_{31}^{0} + R_{31}^{0}\left(R_{11}^{0}\right)^{*} R_{11}^{0}$$

$$= \emptyset + \emptyset\,(\cdots) \quad = \emptyset$$

c) Regular expression for language

$$= R_{13}^{3} = R_{13}^{2} + R_{13}^{2}\left(R_{33}^{2}\right)^{*} R_{33}^{2}$$

$$R_{13}^{2} = R_{13}^{1} + R_{12}^{1}\left(R_{22}^{1}\right)^{*} R_{23}^{1}$$

$$= \emptyset + 1^{*}0\,(\epsilon + 1(1)^{*}0)^{*}\,0$$

$$= 1^{*}0\,(1(1)^{*}0)^{*}\,0$$

$$R_{33}^{2} = R_{33}^{1} + R_{32}^{1}\left(R_{22}^{1}\right)^{*} R_{23}^{1}$$

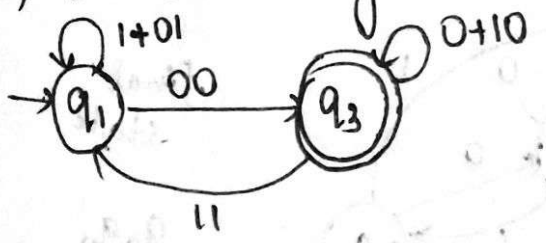
$$= 0 + 1\,(\epsilon + 1(1)^{*}0)^{*}\,0$$

$$= 0 + 1\,(1(1)^{*}0)^{*}\,0$$

$$\therefore R_{13}^{3} = 1^{*}0(1(1)^{*}0)^{*}0 + 0 + 1(1(1)^{*}0)^{*}0$$

$$= 0 + (1^{*}0 + 1)(1(1)^{*}0)^{*}0$$

d) On eliminating state 2



$R = 1 + \epsilon + 01$

$S = 00$

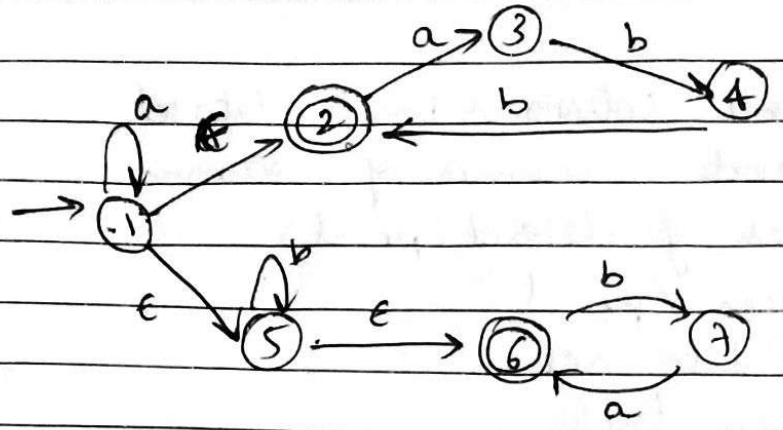
$U = 0 + \epsilon + 10$

$T = 11$

$$R.E = (R + SU^{*}T)^{*}SU^{*} = \left(\left(\begin{matrix}1+\epsilon\\+01\end{matrix}\right) + (00)(0+10+\epsilon)^{*}11\right)^{*}00\,(0+10+\epsilon)^{*}$$

$$= \left((1+01+\epsilon) + 00(0+10)^{*}11\right)^{*}00\,(0+10)^{*}$$

3)



ⓐ find closure of each state

$\epsilon\text{-close}(1) = \{1, 2, 5, 6\}$
$\epsilon\text{-close}(2) = \{2\}$
$\epsilon\text{-close}(3) = \{3\}$
$\epsilon\text{-close}(4) = \{4\}$
$\epsilon\text{-close}(5) = \{5, 6\}$
$\epsilon\text{-close}(6) = \{6\}$
$\epsilon\text{-close}(7) = \{7\}$

ⓑ ε-NFA to DFA

$\epsilon\text{-close}(1) = \{1, 2, 5, 6\} \Rightarrow$ start state
$\delta_d(\{1,2,5,6\}, a) = \epsilon\text{-close}(1, 3) = \{1, 2, 3, 5, 6\}$
$\delta_d(\{1,2,5,6\}, b) = \epsilon\text{-close}(5, 7) = \{5, 6, 7\}$
$\delta_d(\{1,2,3,5,6\}, a) = \epsilon\text{-close}(1, 3) = \{1, 2, 3, 5, 6\}$
$\delta_d(\{1,2,3,5,6\}, b) = \epsilon\text{-close}(4, 5, 7) = \{4, 5, 6, 7\}$
$\delta_d(\{5,6,7\}, a) = \epsilon\text{-close}(6) = 6$
$\delta_d(\{5,6,7\}, b) = \epsilon\text{-close}(5, 7) = \{5, 6, 7\}$
$\delta_d(\{4,5,6,7\}, a) = \epsilon\text{-close}(6) = \{6\}$
$\delta_d(\{4,5,6,7\}, b) = \epsilon\text{-close}(2, 5, 7) = \{2, 5, 6, 7\}$
$\delta_d(6, a) = \phi$
$\delta_d(6, b) = \epsilon\text{-close}(7) = 7$
$\delta_d(\{2,5,6,7\}, a) = \epsilon\text{-close}(3, 6) = \{3, 6\}$
$\delta_d(\{2,5,6,7\}, b) = \epsilon\text{-close}(5, 7) = \{5, 6, 7\}$
$\delta_d(7, a) = 6$
$\delta_d(7, b) = \phi$

$\delta_d(\{3,6\}, a) = \phi$

$\delta_d(\langle 3,6\}, b) = \epsilon\text{-close}(4,7) = \langle 4,7\}$

$\delta_d(\langle 4,7\}, a) = \epsilon\text{-close}(6) = 6$

$\delta_d(\langle 4,7\}, b) = \epsilon\text{-close}(2) = 2$

$\delta_d(2, a) = 3$

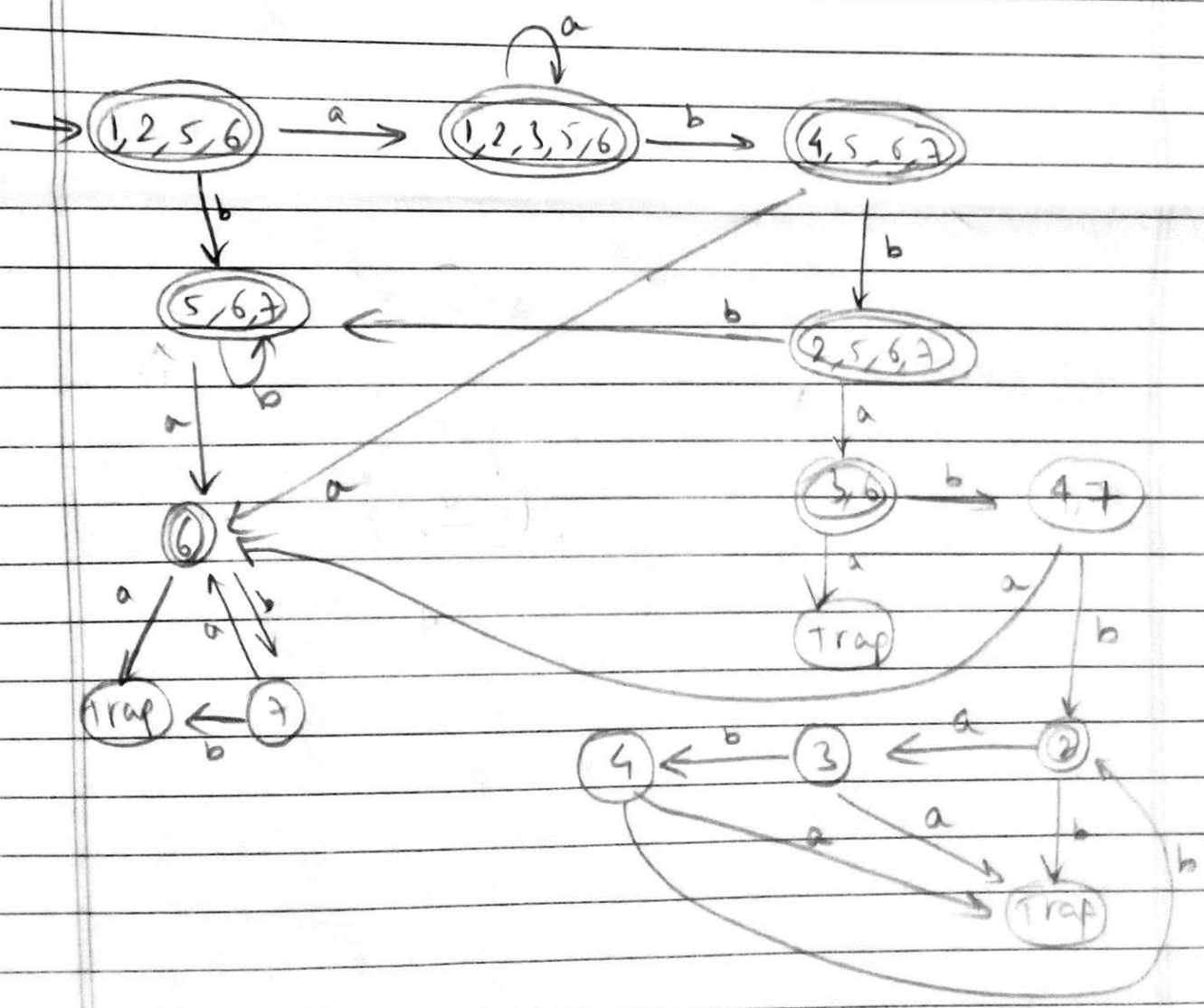
$\delta_d(2, b) = \phi$

$\delta_d(3, a) = \phi$

$\delta_d(3, b) = 4$

$\delta_d(4, a) = \phi$

$\delta_d(4, b) = 2$



4)

a) <u>Pumping Lemma</u>

- It is a property of all regular languages and a technique that is used to show that a given language is not regular.

- <u>Theorem</u> - Let L be a regular language - there exists some constant N, such that for every string $w \in L$, $|w| \geq N$, there exists a way to ~~shee~~ break w into 3 parts,

  $w = xy^{k}z$ such that

  i) $y \neq \epsilon$ ii) $|xy| \leq N$ iii) for all $k \geq 0$, all strings of the form $xy^{k}z \in L$

- This property should hold for all regular languages. N is called 'Pumping Lemma Constant'

Purpose:- To prove some languages cannot be regular

eg:- $L = \{a^n b^n \mid n \geq 0\}$

Proof: i) Let L be regular

ii) so $y \neq \epsilon$, $|y| \geq 1$

Let $M = xy^i z$, $|xy| \leq n$

$i = 0, 1, \ldots$

where $x = a$

$y = a$

$z = a^{n-2} b^n$

Let $i = 0$: $M = a a^i a^{n-2} b^n$

$= a a^{n-2} b^n = a^{n-1} b^n$

We know that $a^{n-1} b^n \notin a^n b^n$

$M \notin L$

$\therefore$ L is not a regular language

$\therefore$ Hence proved by pumping lemma

4.6) Applications of Regular Expression:-

i) In compiler design, it is used for lexical analysis Ex:- To identify and extract tokens (eg:- keywords, identifiers) from code.

ii) Regular expression is used in Data Extraction and Parsing. Ex:- extracting names, addresses/dates from a document

iii) Regular expression is used in network security significantly. Ex:- pattern matching with network traffic logs, to detect common type of malicious payload in network traffic

iv) Command line operations. Ex:- 'grep' commond supports regular exp. to filter & extract specific lines from files

5. * 2 states p & q of a DFA are said to be <u>Indistinguishable</u> if, for all $w \in \Sigma^*$ (w is a string);

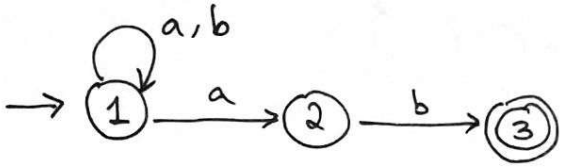
$\delta^*(p, w) \in F$ implies $\delta^*(q, w) \in F$

$\delta^*(p, w) \notin F$ implies $\delta^*(q, w) \notin F$

* 2 states p & q of a DFA are said to be <u>'Distinguishable'</u> by a string 'w' if there exists some string $w \in \Sigma^*$ such that $\delta^*(p, w) \in F$ and $\delta^*(q, w) \notin F$ or vice versa.

Given: NFA

To obtain: Minimised DFA.



$\Sigma = \{a, b\}$

$Q = \{1, 2, 3\}$

δ_N	a	b
→ 1	{1, 2}	{1}
2	ϕ	{3}
* 3	ϕ	ϕ

To DFA by Lazy creation of subsets:

$\delta_D(1, a) = \{1, 2\}$

$\delta_D(1, b) = \{1\}$

$\delta_D(\{1,2\}, a) = \delta_N(1, a) \cup \delta_N(2, a) = \{1, 2\}$

$\delta_D(\{1,2\}, b) = \delta_N(1, b) \cup \delta_N(2, b) = \{1, 3\}$

$\delta_D(\{1,3\}, a) = \{1, 2\}$

$\delta_D(\{1,3\}, b\} = \{1\}$

So,

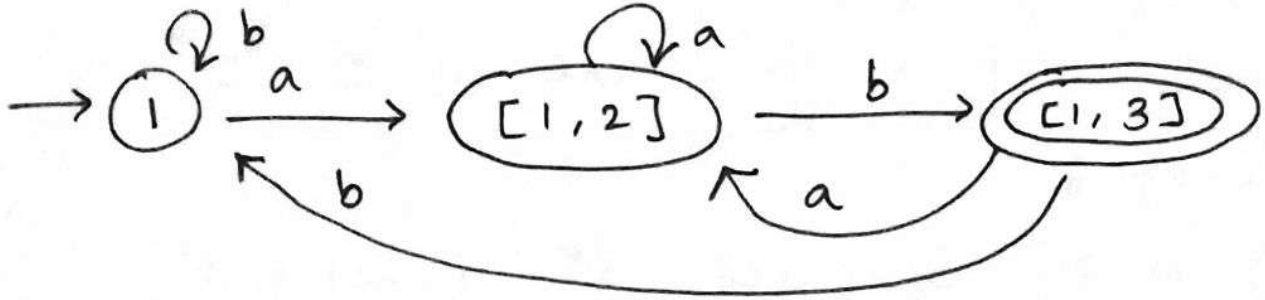
δ_D	a	b
→ 1	{1, 2}	{1}
{1, 2}	{1, 2}	{1, 3}
* {1, 3}	{1, 2}	{1}

So, equivalent DFA:

$Q = \{1, (1,2), (1,3)\}$

$F = \{(1,3)\}$

$q_0 = \{1\}$



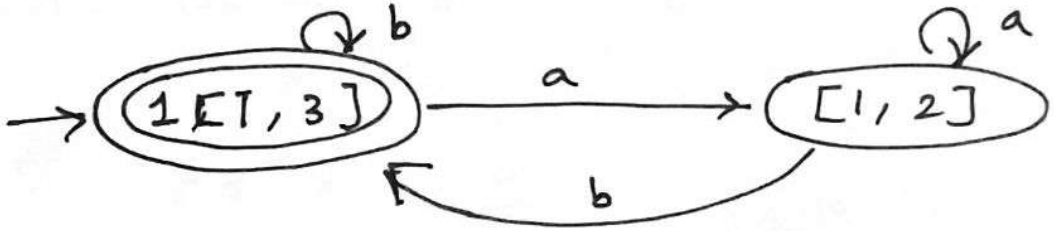
Minimization of DFA:

δ_D	a	b
→ 1	{1, 2}	{1}
{1, 2}	{1, 2}	{1, 3}
* {1, 3}	{1, 2}	{1}

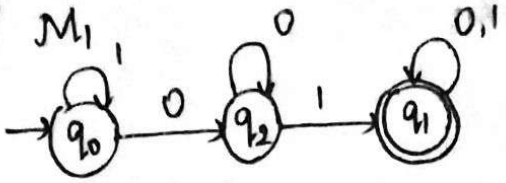
	1	{1,2}	{1,3}
1	=		
{1,2}	×	=	
{1,3}	=	×	=

States '1' & '{1, 3}' are indistinguishable

∴ Minimized DFA:

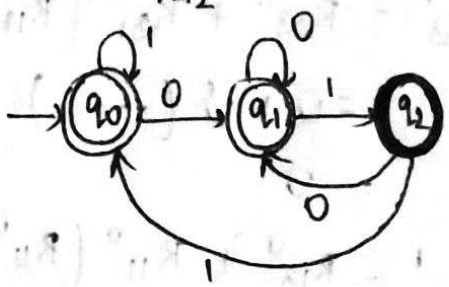


6) M_1



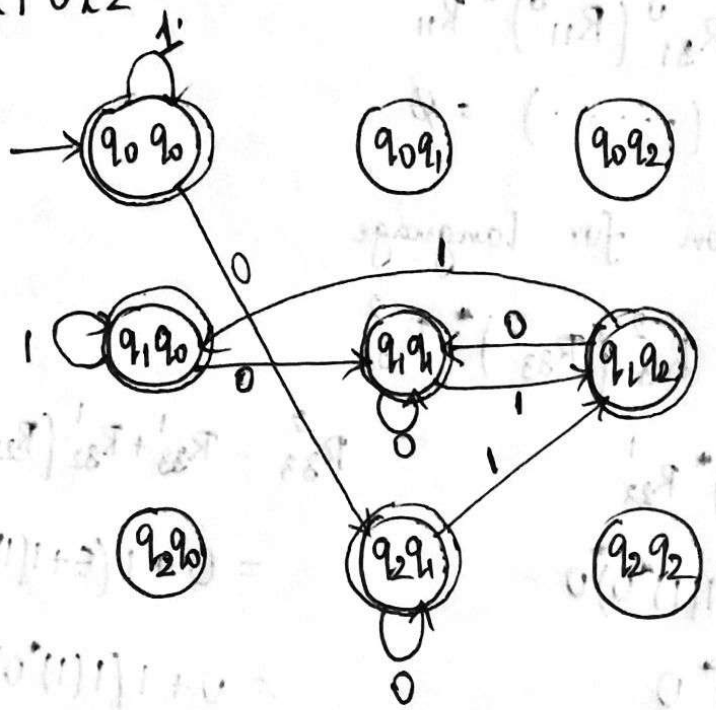
accepts $L1$

M_2



accepts $L2$

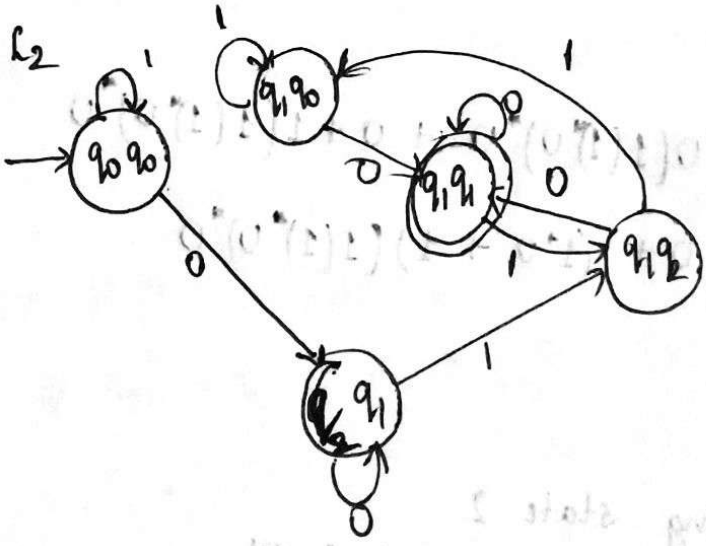
i) $L1 \cup L2$



Final states

q_0q_0
q_1q_0
q_1q_1
q_1q_2
q_2q_1

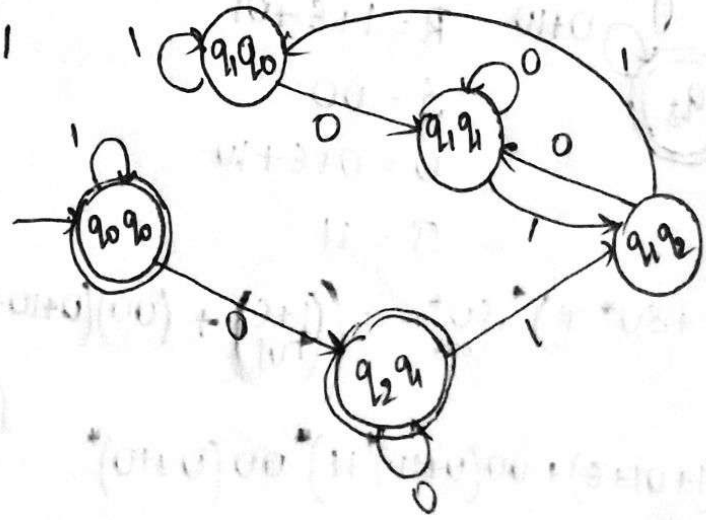
ii) $L_1 \cap L_2$



final state

q_1q_1

iii) $L2 - L1$

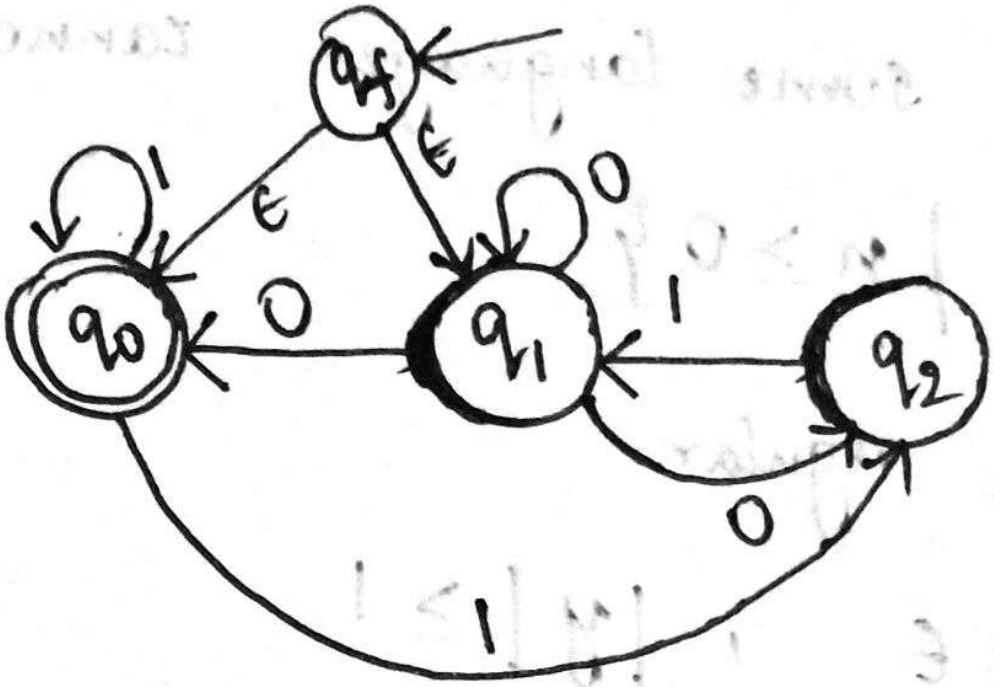
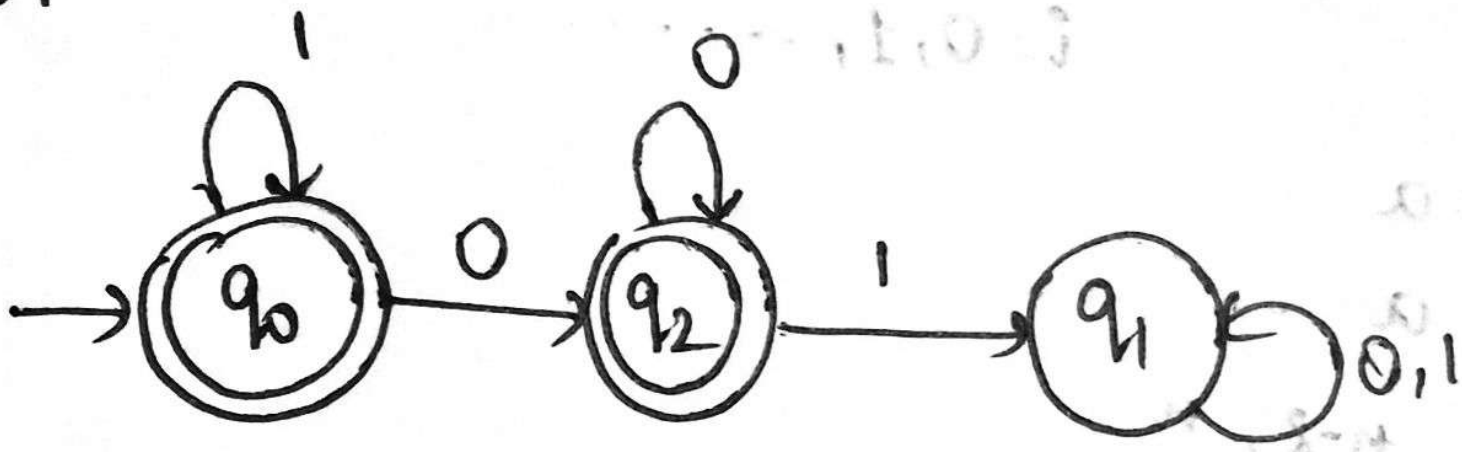


final state

q_0q_0

q_2q_1

iv) L_2^R



v) $\overline{L_1}$